Dr. Lisandro A. Arbilla

Postdoctoral Researcher
LOCEAN-CNRS
Paris, France
lisandro.arbilla@locean.ipsl.fr lisandroarb.github.io

Research Interests

I am a chemical oceanographer specialising in marine biogeochemistry, with a focus on the processes that regulate the ocean carbon cycle and the exchange of CO_2 between the ocean and the atmosphere. My current work focuses on optimising the ocean CO_2 observing system in the Southern Ocean, combining the SOCAT database with autonomous BGC-Argo floats to better estimate the Southern Ocean carbon sink and reduce its uncertainties. In parallel, I collect, process and quality-control underway pCO_2 observations on the Argentine continental shelf and submit them to SOCAT. My previous research addressed the drivers of air-sea CO_2 fluxes and carbonate system variability in the Drake Passage and the northern Antarctic Peninsula.

Academic Appointments

2025-pres. **Postdoctoral Researcher**
LOCEAN-CNRS, Paris, France
Research Topic: Optimising Southern Ocean carbon sink observations using autonomous floats (Horizon Europe project TRICUSO)
Supervisor: Dr. Jean-Baptiste Sallée

2025-26 **CONICET Postdoctoral Fellow**
Instituto Argentino de Oceanografía (IADO-CONICET-UNS), Bahía Blanca, Argentina
Research Topic: Air-sea CO_2 exchange in the Argentine Southwest Atlantic: spatial and temporal variability and its contribution to the global carbon budget
Supervisors: Dr. Carla F. Berghoff, Dr. Celeste López-Abbate

2019-25 **CONICET Doctoral Fellow**
IADO-CONICET-UNS and Universidad de Buenos Aires, Argentina
Research Topic: CO_2 dynamics in the Drake Passage
Supervisors: Dr. Celeste López-Abbate, Dr. Laura A. Ruiz-Etcheverry

Education

2019-25 **Ph.D. in Atmosphere and Ocean Sciences**
Universidad de Buenos Aires (UBA), Argentina
Dissertation: *Dinámica del CO_2 en el Pasaje de Drake* (CO_2 dynamics in the Drake Passage)
Advisors: Dr. Laura A. Ruiz-Etcheverry, Dr. Celeste López-Abbate. Co-advisor: Dr. Lucía C. Kahl

2023-24 **Specialisation in Environmental Engineering**
Universidad Tecnológica Nacional (UTN), Bahía Blanca, Argentina
Final project: *Captura de CO_2 oceánico y el rol de Argentina* (Ocean CO_2 uptake and the role of Argentina)
Advisor: Dr. Carla F. Berghoff

2013-18 **B.Sc. in Oceanography (5-year Licenciatura), Marine Chemistry orientation**
Universidad Nacional del Sur (UNS), Bahía Blanca, Argentina
Thesis: *Trazadores ópticos de materia orgánica disuelta en la zona de El Rincón-Talud* (Optical tracers of dissolved organic matter)
Advisors: Dr. Ana M. Martínez, Dr. John E. Garzón-Cardona

Peer-Reviewed Publications

In review:

[1] **Arbilla, L.A.**, Berghoff, C.F., Ruiz-Etcheverry, L.A., del Valle, D.A., Segura, V., Tonini, M.H., Veccia, M.H. Spatial variability of air-sea CO_2 fluxes in the Northern Patagonian Shelf during summer. Submitted to *Journal of Marine Systems*.

[2] **Arbilla, L.A.**, López-Abbate, M.C., Ruiz-Etcheverry, L.A., Berden, G., Gilabert, A.S., Kahl, L.C. From climate variability to compound events: ENSO-SAM interactions regulate air-sea CO_2 exchange in the Drake Passage. Submitted to *JGR: Biogeosciences*.

[3] Delaigue, L., Landschützer, P., Wimart-Rousseau, C., **Arbilla, L.A.**, Sallée, J.-B., et al. The Southern Ocean CO_2 sink in an evolving observing system. Submitted to *Global Biogeochemical Cycles*. Preprint: 10.22541/essoar.15002449/v1

[4] Gilabert, A.S., López-Abbate, M.C., **Arbilla, L.A.**, et al. Physical-biological interactions create heterogeneous sea-air CO_2 fluxes on the northern Patagonian Shelf during the low-productivity season. Submitted to *Limnology and Oceanography*.

Published:

[1] Álvarez, M., García-Ibáñez, M.I., Acerbi, R., Santiago-Domenech, R., Guallart, E.F., Fajar, N.M., **Arbilla, L.A.**, 2026. Ocean acidification at the crossroads I: Harmonizing unpurified and purified meta-cresol purple spectrophotometric pH_T measurements based on absorbance data. *Limnology and Oceanography: Methods*, 24, e70023, 10.1002/lom3.70023

[2] Garzón-Cardona, J.E., Martínez, A.M., Alonso, C., Palma, E.D., González, M.B., Barrera, F.M., **Arbilla, L.A.**, et al., 2026. Insight into origin and transformation of dissolved organic matter under contrasting nutrient conditions on the Northern Patagonian Continental Shelf, Southwestern Atlantic Ocean. *Marine Environmental Research*, 213, 107660, 10.1016/j.marenvres.2025.107660

[3] **Arbilla, L.A.**, López-Abbate, M.C., Ruiz-Etcheverry, L.A., Gilabert, A.S., Kahl, L.C., Osiroff, A.P., 2025. Regionally distinct drivers of the carbonate system dynamics in the Drake Passage and northern Antarctic Peninsula. *Journal of Marine Systems*, 249, 104070, 10.1016/j.jmarsys.2025.104070

[4] Kitch, G.D., Duke, P.J., Grabb, K.C., Simancas-Giraldo, S., Adekunbi, F.O., Addey, C.I., **Arbilla, L.A.**, et al., 2025. Early career recommendations for the equitable growth of a marine carbon dioxide removal sector. *Perspectives of Earth and Space Scientists*, 6, e2024CN000246, 10.1029/2024CN000246

[5] Gilabert, A.S., López-Abbate, M.C., Flombaum, P., Unrein, F., **Arbilla, L.A.**, et al., 2025. Planktonic drivers of carbon transformation during different stages of the spring bloom at the Patagonian Shelf-break front, Southwestern Atlantic Ocean. *Biogeochemistry*, 168, 10, 10.1007/s10533-024-01192-6

[6] **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., López-Abbate, M.C., Kahl, L.C., 2024. CO_2 sink and source zones delimited by marine fronts in the Drake Passage. *Progress in Oceanography*, 223, 103246, 10.1016/j.pocean.2024.103246

Data Products and Reports

Data products:

[1] Gilabert, A.S., López-Abbate, M.C., Garzón-Cardona, J.E., **Arbilla, L.A.**, et al., 2026. Hydrographic conditions and fluorescence peaks of dissolved organic matter along a transect at the Brazil-Malvinas Confluence. Zenodo, zenodo.org/records/18331411

[2] Bakker, D.C.E., et al. (incl. **Arbilla, L.A.**), 2026. Surface Ocean CO_2 Atlas Database Version 2026 (SOCATv2026) (NCEI Accession 0315110). NOAA NCEI, 10.25921/8dba-fr90

[3] Bakker, D.C.E., et al. (incl. **Arbilla, L.A.**), 2025. Surface Ocean CO_2 Atlas Database Version 2025 (SOCATv2025) (NCEI Accession 0304549). NOAA NCEI, 10.25921/648f-fv35

[4] Álvarez, M., García-Ibáñez, M.I., Acerbi, R., Santiago-Domenech, R., Guallart, E.F., Fajar, N.M., **Arbilla, L.A.**, 2025. Unpur-Pur data compilation. DIGITAL.CSIC,

10.20350/DIGITALCSIC/17516

[5] Gilabert, A.S., López-Abbate, M.C., **Arbilla, L.A.**, et al., 2025. Physical-biological interactions regulate sea-air CO_2 fluxes on the northern Patagonian shelf during the low-productivity season. Zenodo, zenodo.org/records/17055333

[6] Osiroff, A.P., Kahl, L.C., **Arbilla, L.A.**, 2025. ARGAU cruise data: carbonate system measurements (2003-2004). Zenodo, zenodo.org/records/15212764

[7] Bakker, D.C.E., et al. (incl. **Arbilla, L.A.**), 2024. Surface Ocean CO_2 Atlas Database Version 2024 (SOCATv2024) (NCEI Accession 0293257). NOAA NCEI, 10.25921/9wpn-th28

[8] Berghoff, C.F., Veccia, M.H., **Arbilla, L.A.**, 2024. Partial pressure of carbon dioxide, water temperature, salinity and other variables collected from surface underway observations during the R/V Víctor Angelescu cruises in the South Atlantic Ocean in 2023 (NCEI Accession 0286984). NOAA NCEI, 10.25921/fvak-g381

[9] Data submission to the UNESCO-IOC SDG 14.3.1 Data Portal (oa.iode.org): pH and total alkalinity from two COSTAL-AR time-series cruises (VA2021-01 and VA2021-06).

Reports:

[1] Epherra, L., Berghoff, C.F., Luz Clara, M., **Arbilla, L.A.**, et al., 2025. Sistema de carbonatos en la serie de tiempo COSTAL-AR: resultados de las campañas VA2021-01 y VA2021-06. INIDEP Research Report 084/25, 18 pp.

[2] Berghoff, C.F., **Arbilla, L.A.**, Veccia, M.H., 2025. Sistema de medición en continuo de la pCO_2 superficial del BIPO Víctor Angelescu: procesamiento de la información adquirida durante 2025 y aporte de datos a SOCAT. INIDEP Advisory Report 113/2025, 6 pp.

[3] Berghoff, C.F., **Arbilla, L.A.**, 2025. Sistema de medición en continuo de la pCO_2 superficial del BIPO Víctor Angelescu: procesamiento de la información adquirida durante 2024 y aporte de datos a SOCAT. INIDEP Advisory Report 003-25, 6 pp.

[4] Berghoff, C.F., **Arbilla, L.A.**, del Valle, D.A., et al., 2024. Variabilidad espacial de los flujos de CO_2 mar-atmósfera en la Plataforma Norpatagónica durante el verano de 2024. INIDEP Research Report 047-24, 28 pp.

[5] Berghoff, C.F., Veccia, M.H., **Arbilla, L.A.**, 2024. Sistema de medición en continuo de la pCO_2 superficial del BIPO Víctor Angelescu: procesamiento de la información adquirida durante 2022-2023 y aporte de datos a SOCAT. INIDEP Advisory Report 023-24, 6 pp.

[6] Álvarez, M., Acerbi, R., García-Ibáñez, M.I., **Arbilla, L.A.**, et al., 2023. RADPROF 2023 oceanographic cruise report. Instituto Español de Oceanografía (IEO-CSIC), 45 pp., NCEI 0291177

[7] Piola, A.R., **Arbilla, L.A.**, Baqués, M., et al., 2022. Campaña Oceanográfica Austral 2022-03 "SAM-20". Cruise report, 55 pp., 10.13140/RG.2.2.20250.25280

[8] Romero, S.I., Dávila, P.M., **Arbilla, L.A.**, Osiroff, A.P., 2020. Definición de línea de convergencia subtropical para cartas náuticas H-1 y H-2. Servicio de Hidrografía Naval, Argentina.

Research Projects

Current:

2025-28 **TRICUSO: Three Research Infrastructures together: Carbon Uptake Southern Ocean**
Horizon Europe, Grant No. 101188028. Total budget: €10,000,000. PI: Dr. Richard Sanders (NORCE)
Role: Postdoctoral researcher in WP3 (Observing system design and strategy) and WP5 (Optimisation of the Southern Ocean carbon observing system)

2026-28 **Establishing baseline conditions for responsible mCDR deployment in the Argentine Southwestern Atlantic**
ClimateWorks Foundation. Budget: USD 100,000. PI: Dr. Carla F. Berghoff (INIDEP)
Role: Researcher, mainly in WP1 (Air-sea CO_2 fluxes and carbonate system pattern recognition), and partially in WP2 and WP4

2026-28 **Profundizando en los mecanismos de captura y destino del carbono en la plataforma continental norte de Argentina**
CONICET PIP 11220250101386CO. Budget: ARS 20,000,000. PI: Dr. Celeste López-Abbate (IADO)
Role: Researcher, air-sea CO_2 fluxes and their physical and biological drivers

2026-29 **Génesis y transformación de la materia orgánica y los nutrientes en el margen continental argentino: influencia del aporte continental y de las masas de agua**
PGI 24/ZG40 (SGCyT-UNS). Budget: ARS 1,500,000. PI: Dr. John E. Garzón-Cardona
Role: Team member

2026-27 **Rutas microbianas involucradas en la transformación y el secuestro de carbono en el Atlántico Sudoccidental**
PGI 24/ZG41 (SGCyT-UNS). Budget: ARS 1,500,000. PI: Dr. Azul S. Gilabert
Role: Team member

2023-pres. **Smart-Port Net-Zero Challenge: natural decarbonisation potential of the Bahía Blanca Estuary**
Port of Bahía Blanca (winning proposal, 2022). Budget: ARS 1,500,000. PIs: Dr. Celeste López-Abbate, Dr. Ana L. Delgado (IADO)
Role: Carbonate system measurements and management and analysis of CO_2 flux data

Previous:

2024-25 PGI 24/ZQ24 (SGCyT-UNS): Materia orgánica y su influencia en el transporte y biodisponibilidad de oligoelementos en ambientes de transición. Budget: ARS 800,000. PI: Dr. John E. Garzón-Cardona

2023-25 PGI 24/Q126 (SGCyT-UNS): Dinámica de la materia orgánica disuelta: vinculación de las propiedades ópticas y químicas en ambientes acuáticos y de transición. Budget: ARS 80,000. PI: Dr. Ana M. Martínez

2020-25 COASTCARB: Carbon balance for South Patagonian and Antarctic coastal ecosystems in times of rapid glacier melt. H2020 MSCA-RISE No. 872690. Total budget: €1,400,000. PI: Dr. Kerstin Jerosch (AWI)

2022-25 PICT-2020-SERIEA-02309: Variabilidad espacio-temporal de los flujos de CO_2 en la Plataforma Continental Norpatagónica. Budget: ARS 1,410,000. PI: Dr. Ana L. Delgado

2022-24 BioMMAr: Biodiversidad del plancton microbiano del Mar Argentino y su rol en la trama trófica y los flujos de carbono. Pampa Azul (MINCyT). PI: Dr. Pedro Flombaum

2021-24 PIP 11220200102681CO (CONICET): Vías microbianas de secuestro de carbono en la Plataforma Norpatagónica. Budget: ARS 1,825,000. PI: Dr. Celeste López-Abbate

2019-20 PGI 24/Q104 (SGCyT-UNS): Evaluación de las propiedades ópticas de la materia orgánica disuelta. Budget: ARS 50,000. PI: Dr. Ana M. Martínez

Oceanographic Cruises and Fieldwork

2024-pres. **Underway pCO_2 system, R/V Víctor Angelescu (INIDEP)**
Maintenance and quality control of the General Oceanics 8050 system; validation and processing of data from 16 national cruises and submission to SOCAT

2023 **RADCAN-RADPROF 2023-LANDERFLEET cruises, R/V Ramón Margalef (IEO-CSIC, Spain)**
21 days at sea. Chief scientist: Dr. Marta Álvarez

2022 **South Atlantic Meridional Overturning Circulation (SAMOC) 2022, ARA Austral (Argentina)**
12 days at sea, 1,685 nautical miles. Chief scientist: Dr. Alberto Piola

2019, 2023 **Coastal surveys in the Bahía Blanca Estuary, Argentina**

Research Stays and Training

2024 **INIDEP, Mar del Plata, Argentina**
Training in open-cell potentiometric total alkalinity measurements. Hosts: Dr. Carla F. Berghoff, Dr. Lucía Epherra

2023-24 **INIDEP (virtual, 100 h)**
Processing and quality control of underway pCO_2 data (GO-8050, pCO_2_sys) following SOCAT/WOCE procedures. Host: Dr. Carla F. Berghoff

2023 **Instituto Español de Oceanografía (IEO-CSIC), A Coruña, Spain (2.5 months, POGO fellowship)**
Chemical oceanography training (dissolved oxygen, pH, total alkalinity, DIC, nutrients) at the INOCEN laboratory and on board. Host: Dr. Marta Álvarez

2022 **Centro de Investigación en Ecosistemas de la Patagonia (CIEP), Chile**
Training in carbonate system measurements. Host: Dr. Rodrigo Torres

2017-18 **Environmental Chemistry Laboratory (UNS-INQUISUR), Bahía Blanca, Argentina**
Fluorescence and absorption spectroscopy of photosynthetic pigments; dissolved carbohydrate analyses. Host: Dr. Ana M. Martínez

Fellowships and Awards

2025-28 CONICET Postdoctoral Fellowship

2023 POGO Fellowship for a research stay at IEO-CSIC, A Coruña, Spain

2023 NOAA travel grant: Workshop on surface ocean pCO_2 observations, synthesis and data products (VLIZ, Oostende)

2022 Smart-Port Net-Zero Challenge, winning team (Port of Bahía Blanca)

2022 POGO grant: Best practices for biogeochemical ocean observation (POGO-SAMS)

2022 Registration grant, 5th International Symposium on the Ocean in a High-CO_2 World (CIMA-IFAECI)

2021 IAEA grant: Regional training on carbonate system measurements for SDG indicator 14.3.1

2021 INVEMAR-UNESCO/IOC grant: Climate change, blue carbon and ecosystem-based adaptation

2019-25 CONICET Doctoral Fellowship

Teaching Experience

2024-pres. **Teaching Assistant, Climatology**
Department of Geography and Tourism, Universidad Nacional del Sur (competitive position)

2025 **Teaching Assistant, Field Practice II**
Oceanography Technician degree, Universidad Nacional del Sur

2024 **Teaching Assistant, Field Practice I**
Oceanography Technician degree, Universidad Nacional del Sur

Presentations

International:

[1] Berghoff, C.F., **Arbilla, L.A.**, Basañes, S., et al., 2026. Progress in ocean acidification observing capacity in the Argentine Atlantic. 6th International Symposium on the Ocean in a High-CO_2 World, Wellington, New Zealand.

[2] **Arbilla, L.A.**, Sallée, J.-B., Landschützer, P., et al., 2026. BGC-Argo floats and SOCAT as a pCO_2 observing system in the Southern Ocean: a multi-model subsampling assessment. 6th International Symposium on the Ocean in a High-CO_2 World, Wellington, New Zealand.

[3] **Arbilla, L.A.**, Berghoff, C.F., Ruiz-Etcheverry, L.A., et al., 2026. Seasonal variability of air-sea

CO_2 fluxes in the Northern Patagonian Shelf. 6th International Symposium on the Ocean in a High-CO_2 World, Wellington, New Zealand.

[4] Basañes, S., Berghoff, C.F., **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., 2026. Empirical models of total alkalinity on the Northern Continental Shelf for a regional estimation of the carbonate system. 6th International Symposium on the Ocean in a High-CO_2 World, Wellington, New Zealand.

[5] **Arbilla, L.A.**, Sallée, J.-B., Landschützer, P., et al., 2026. A multi-model subsampling framework to evaluate BGC-Argo floats and SOCAT as a pCO_2 observing system in the Southern Ocean. Optical Ocean Carbon Observing Design Workshop, VLIZ, Ostend, Belgium.

[6] **Arbilla, L.A.**, Berghoff, C.F., Ruiz-Etcheverry, L.A., et al., 2026. The CO_2 sink in a fisheries-relevant region of the Argentine shelf break. Ocean Sciences Meeting 2026, Glasgow, UK. Oral presentation.

[7] Sallée, J.-B., et al., **Arbilla, L.A.**, 2026. Southern Ocean regime shift revealed by long-term repeated hydrographic observations in East Antarctica. Ocean Sciences Meeting 2026, Glasgow, UK.

[8] Berghoff, C.F., Epherra, L., Basañes, S., **Arbilla, L.A.**, et al., 2025. Comprendiendo la dinámica del CO_2 en la Zona Común de Pesca: avances y perspectivas. Workshop of the Binational Environmental Monitoring Program of the ZCP 2023-2028, Mar del Plata, Argentina.

[9] Álvarez, M., García-Ibáñez, M.I., Acerbi, R., Santiago-Domenech, R., Guallart, E.F., Fajar, N.M., **Arbilla, L.A.**, Hassoun, A.E.R., 2025. Coherent spectrophotometric pH measurements: the purified and unpurified m-cresol purple dilemma. WEPAL-QUASIMEME Workshop on Quality Assurance for Inorganic Carbon Measurements, National Oceanography Centre, Southampton, UK.

[10] Gilabert, A.S., López-Abbate, M.C., **Arbilla, L.A.**, et al., 2024. Explorando el potencial de secuestro de carbono en la Plataforma Patagónica a través de la actividad del plancton microbiano. IV Jornadas Internacionales de Ambiente, Viedma, Argentina. Extended abstract.

[11] Gilabert, A.S., López-Abbate, M.C., **Arbilla, L.A.**, et al., 2024. Descubriendo la vida microscópica del Frente de Talud Patagónico: fitoplancton, bacterias y virus. IV Jornadas Internacionales de Ambiente, Viedma, Argentina. Extended abstract.

[12] **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., López-Abbate, M.C., 2023. Detection of dynamic fronts using satellite altimetry. Ocean Surface Topography Science Team Meeting (OSTST), Puerto Rico, USA. Extended abstract, 10.24400/527896/a03-2023.3854

[13] **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., López-Abbate, M.C., Kahl, L.C., 2023. CO_2 fluxes in the Drake Passage: a comprehensive exploration. Ocean Acidification Week, LAOCA session, GOA-ON (online). Oral presentation.

[14] López-Abbate, M.C., **Arbilla, L.A.**, Gilabert, A.S., et al., 2023. ENECO$_2$: an initiative to identify natural decarbonization pathways in Argentina. PICES 2023 Annual Meeting, Seattle, USA.

[15] Gilabert, A.S., **Arbilla, L.A.**, López-Abbate, M.C., et al., 2023. Interacción del sistema científico-tecnológico y áreas de gestión integrada del Estuario de Bahía Blanca para la implementación eficiente de herramientas de emisión negativa de CO_2. III Congreso Iberoamericano de Gestión Integrada de Áreas Litorales (GIAL), Mar del Plata, Argentina.

[16] **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., López-Abbate, M.C., Kahl, L.C., Bianchi, A.A., 2022. CO_2 sink and source zones delimited by marine fronts in the Drake Passage. 5th International Symposium on the Ocean in a High-CO_2 World, Lima, Peru. Oral presentation.

National:

[1] Epherra, L., Berghoff, C.F., Luz Clara, M., **Arbilla, L.A.**, et al., 2025. Variabilidad del estado de saturación de la aragonita en la serie de tiempo ecológica marina COSTAL-AR. XII Jornadas Nacionales de Ciencias del Mar, Puerto Madryn, Argentina.

[2] Berghoff, C.F., **Arbilla, L.A.**, del Valle, D.A., et al., 2025. Variabilidad espacial de los flujos de CO_2 mar-atmósfera en la Plataforma Norpatagónica durante el verano de 2024. XII Jornadas Nacionales de Ciencias del Mar, Puerto Madryn, Argentina.

[3] Gilabert, A.S., López-Abbate, M.C., **Arbilla, L.A.**, et al., 2025. Influencia de la comunidad

microbiana planctónica sobre la materia orgánica disuelta y el intercambio de CO_2 mar-atmósfera en una época productiva reducida. XII Jornadas Nacionales de Ciencias del Mar, Puerto Madryn, Argentina.

[4] López-Abbate, M.C., Gilabert, A.S., **Arbilla, L.A.**, et al., 2025. Estructura y fenología del plancton microbiano: dos atributos clave en la transformación del carbono en la Plataforma Continental Argentina. XII Jornadas Nacionales de Ciencias del Mar, Puerto Madryn, Argentina.

[5] **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., López-Abbate, M.C., Kahl, L.C., Bianchi, A.A., 2022. Zonas de captura y emisión de CO_2 en el Pasaje de Drake. XI Jornadas Nacionales de Ciencias del Mar, Comodoro Rivadavia, Argentina. Oral presentation.

[6] **Arbilla, L.A.**, Ruiz-Etcheverry, L.A., López-Abbate, M.C., 2022. Detección de frentes dinámicos con altimetría satelital. XI Jornadas Nacionales de Ciencias del Mar, Comodoro Rivadavia, Argentina.

[7] **Arbilla, L.A.**, Garzón-Cardona, J.E., Martínez, A.M., Lara, R.J., 2019. Señales fluorescentes de la materia orgánica disuelta en El Rincón y frente del talud. V Reunión Argentina de Geoquímica de la Superficie, La Plata, Argentina.

[8] **Arbilla, L.A.**, Gilabert, A.S., Garzón-Cardona, J.E., Martínez, A.M., Lara, R.J., 2018. Distribución de la materia orgánica disuelta cromofórica en El Rincón y frente del talud. X Jornadas Nacionales de Ciencias del Mar, Buenos Aires, Argentina.

[9] **Arbilla, L.A.**, Garzón-Cardona, J.E., Martínez, A.M., Lara, R.J., 2018. Trazadores ópticos de la materia orgánica disuelta en El Rincón y frente del talud. X Jornadas Nacionales de Ciencias del Mar, Buenos Aires, Argentina.

Academic Service and Outreach

2026 **SOCAT Quality Control Hackathon**
Quality control of surface ocean CO_2 data for the global SOCAT database

2025-pres. **Curriculum Committee member, Oceanography Technician degree, UNS**

2025-27 **Alternate fellows representative, Board of Directors, IADO**

2024-pres. **Peer reviewer**
Journal of Geophysical Research: Oceans, *Journal of Marine Systems*, *Marine Chemistry*

2024 **Radio interview on oceanographic research**
Récord Puan Radios and Cuarto Poder

2023-pres. **Outreach project "Un Mar de Preguntas"**
IADO-CONICET, Bahía Blanca. Coordinator: Dr. Celeste López-Abbate

2023 **Argentine Science Week (XXI edition)**
Ocean science activities for primary and secondary schools

2016-19 **Curriculum Committee member, Licentiate in Oceanography, UNS**

Scientific networks:

2023-pres. NF-POGO Alumni Network for the Ocean (NANO)

2022-pres. International Carbon Ocean Network for Early Career (ICONEC)

2022-pres. Global Ocean Acidification Observing Network (GOA-ON)

2019-pres. Latin-American Ocean Acidification Network (LAOCA)

Professional Development

2026 Workshop: Underway pCO_2 measurements on ships: the General Oceanics systems and data processing. NOAA GOMO (Drs. Denis Pierrot and Leticia Barbero)

2023 Workshop on surface ocean pCO_2 observations, synthesis and data products. VLIZ, Oostende, Belgium (online)

2022	Best practices for biogeochemical ocean observation: instrumentation, operation, quality control. POGO and SAMS
2022	Standards of Training, Certification and Watchkeeping for Seafarers (STCW'78). Prefectura Naval Argentina
2021	Regional training on carbonate system measurements for SDG indicator 14.3.1 (60 h). IAEA and INVEMAR
2021	Climate change: blue carbon and ecosystem-based adaptation in marine and coastal ecosystems (60 h). INVEMAR and UNESCO/IOC
2019-20	Graduate courses: Climate change fundamentals; Climatology; Ecology of marine fronts; Oceanography seminars; Satellite oceanography (OFISAT); Estuarine geomorphology and dynamics

Skills and Languages

Computing	Python, R, MATLAB, Ocean Data View, Surfer
Laboratory	Carbonate system (DIC, total alkalinity, spectrophotometric pH), dissolved oxygen, nutrients, dissolved organic matter (CDOM, FDOM), chlorophyll-a; underway pCO_2 systems
Languages	Spanish (native), English (upper-intermediate, B2)